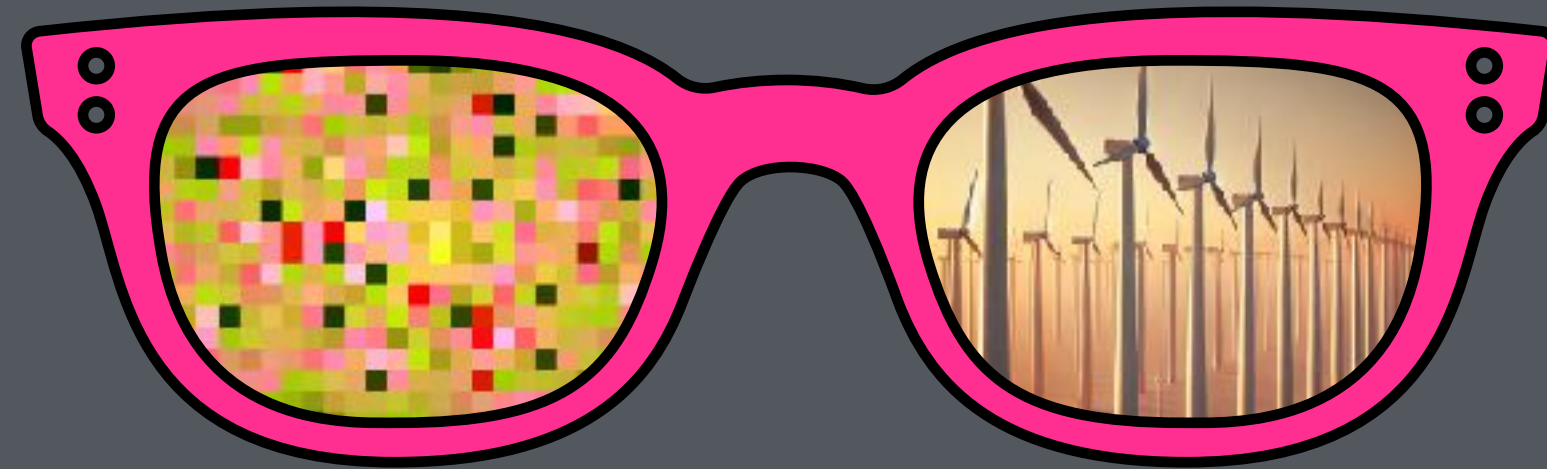


# Data and the State

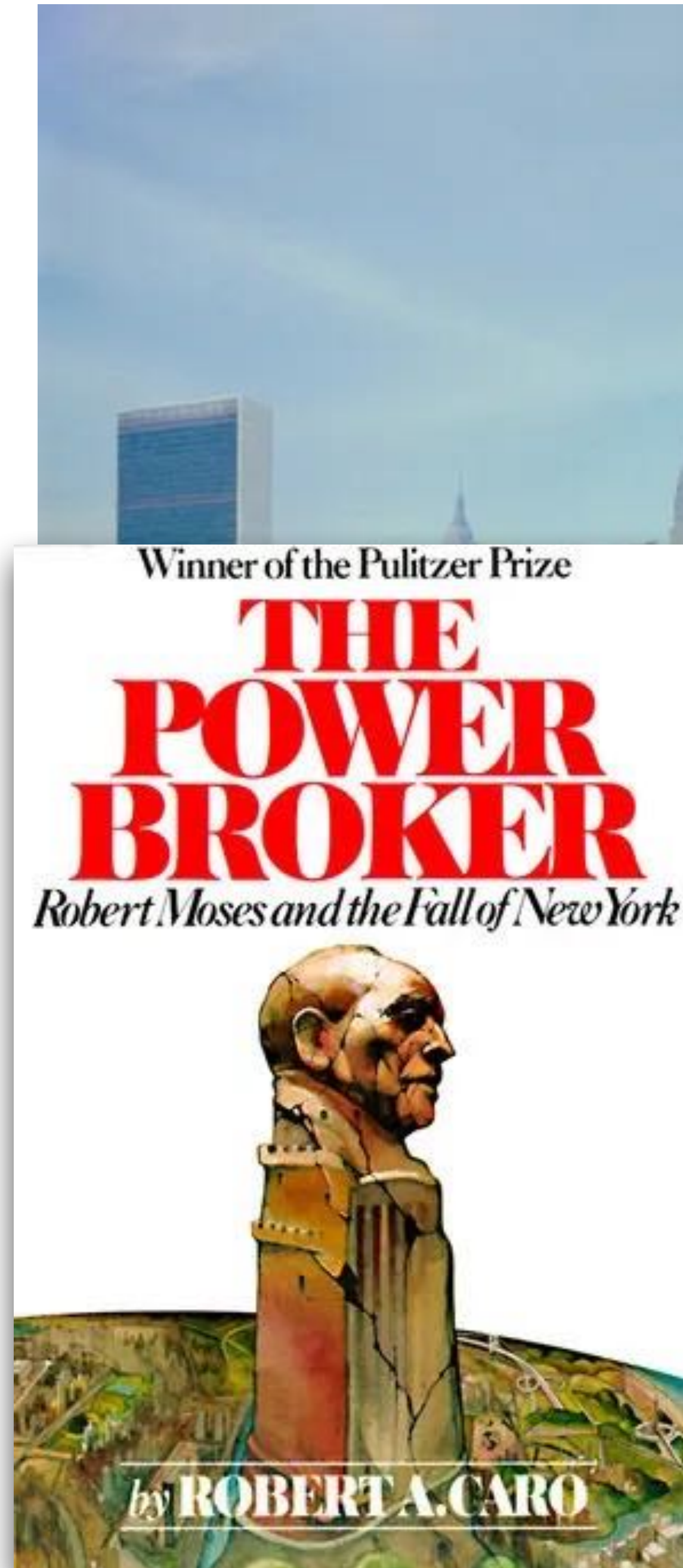
DATA 20195 / 30195



Neighborhoods  
and policy power

Lecture 17, Monday May 18





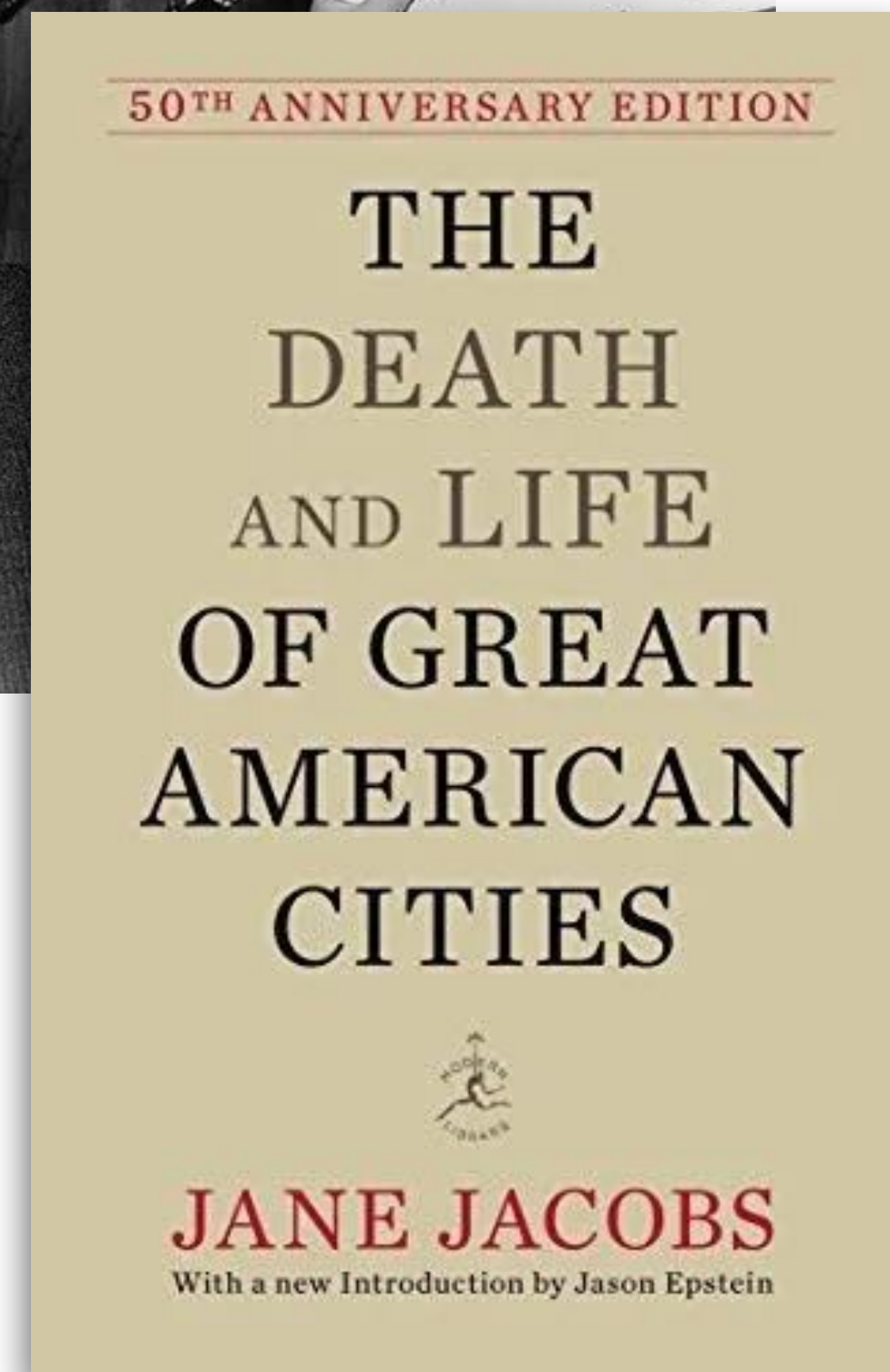
**Robert Moses**  
1888-1981

*infrastructure czar*



**Jane Jacobs**  
1916-2006

*community advocate*





# Moses as power broker

- Long Island State Park Commission (President, 1924–1963)
- New York State Council of Parks (Chairman, 1924–1963)
- New York Secretary of State (1927–1929)
- Bethpage State Park Authority (President, 1933–1963)
- Emergency Public Works Commission (Chairman, 1933–1934)
- Jones Beach Parkway Authority (President, 1933–1963)
- New York City Department of Parks (Commissioner, 1934–1960)
- Triborough Bridge and Tunnel Authority (Chairman, 1934–1968)
- New York City Planning Commission (Commissioner, 1942–1960)
- New York State Power Authority (Chairman, 1954–1962)
- New York's World Fair (President, 1960–1966)
- Office of the Governor of New York (Special Advisor on Housing, 1974–1975)
- plus tons of informal or quasi-formal roles, like “NYC Construction Coordinator” (1946)

**pools,  
bridges,  
tunnels,  
highways**

power grew with  
every mayoral shift  
in mid-century

for many years, had  
sole authority to  
negotiate for NYC  
projects in DC



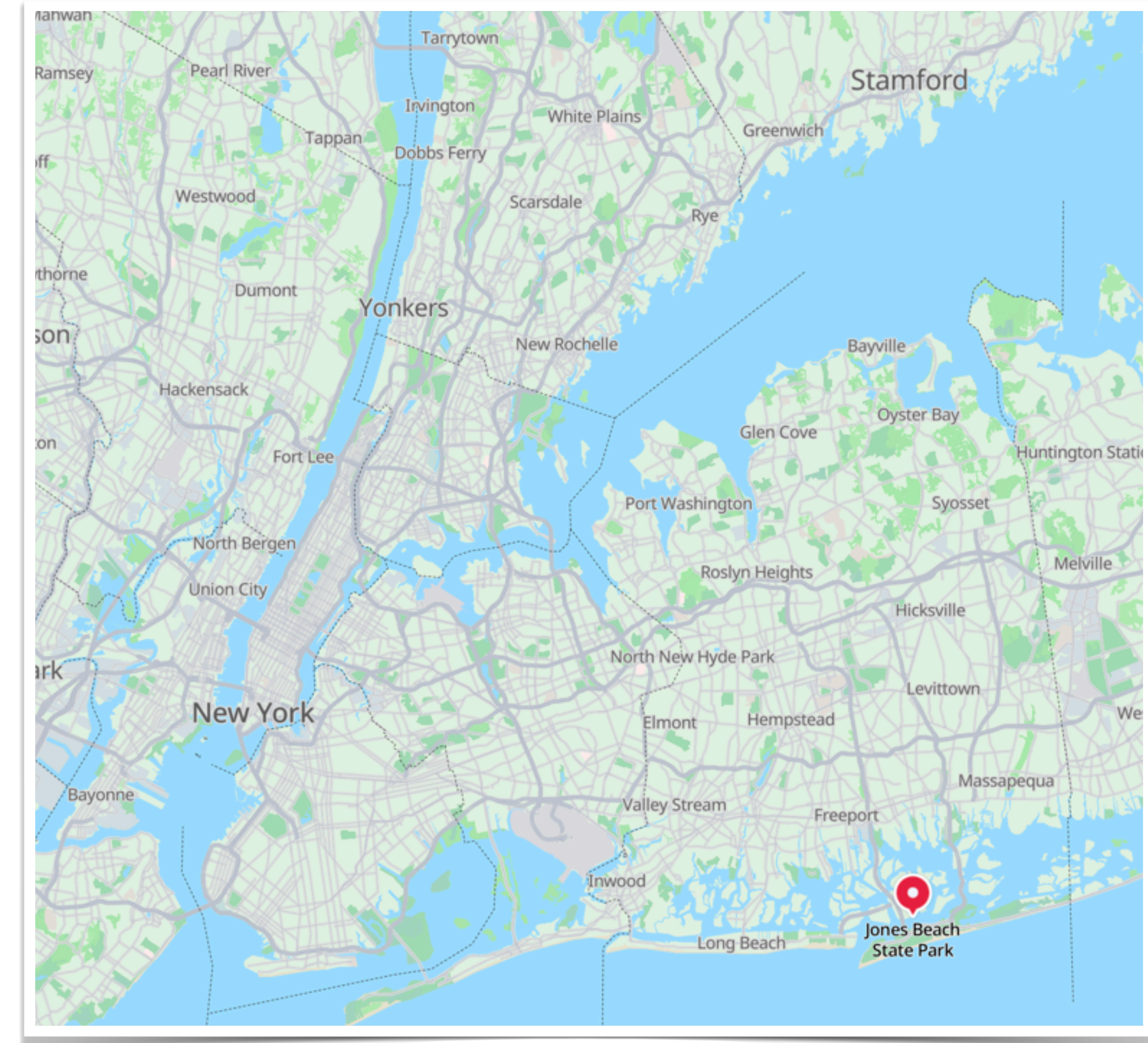
# Moses legacy

Moses response to critics: "I raise my stein to the builder who can remove ghettos without moving people as I hail the chef who can make omelets without breaking eggs."

I have often said that  
another biography of Robert  
Moses could be written that  
would be titled:  
**"At Least He Got it Built."**



## Jones Beach controversy





# Counterpoint with Jacobs

- Advocated for an unsentimental view of neighborhoods as **power centers** that can influence policy decisions, and argued that neighborhoods of 30,000-50,000 are not “useful”
- Only three scales matter — the street neighborhood (hyperlocal), the whole city (global), and something in between that she calls the **district**, needing about 100,000 people
- She doesn’t mean electoral districts! She means a powerful nexus of community with “localized self-government,” “natural cross-connections,” and a network of about 100 leaders who represent different special-interest groups
- “Our cities possess many islandlike neighborhoods too small to work as districts, and these include not only the project neighborhoods inflicted by planning, but also many unplanned neighborhoods. These unplanned, too small units have grown up historically, and often are enclaves of distinctive ethnic groups.”

# Jacobs, continued

- Special-interest groups to include “churches, P-TA’s [parent/teacher associations], businessmen's associations, political clubs, local civic leagues, fund-raising committees for health campaigns or other public causes, sons of such-and-such a village (common clubs among Puerto Ricans today, as they have been with Italians), property owners' associations, block improvement associations, protesters against injustices, and so on, ad infinitum.”
- Strength in organizing would come through “hop-and-skip” relationships
- Resilient networks can form hilariously specific ad hoc efforts, e.g., **“Committee of Neighbors to Get the Clock on Jefferson Market Courthouse Started”**



*arrested in 1968 for  
disrupting a public  
meeting, protesting  
Moses project to put a  
10-lane highway through  
lower Manhattan*





what forces make  
neighborhoods?



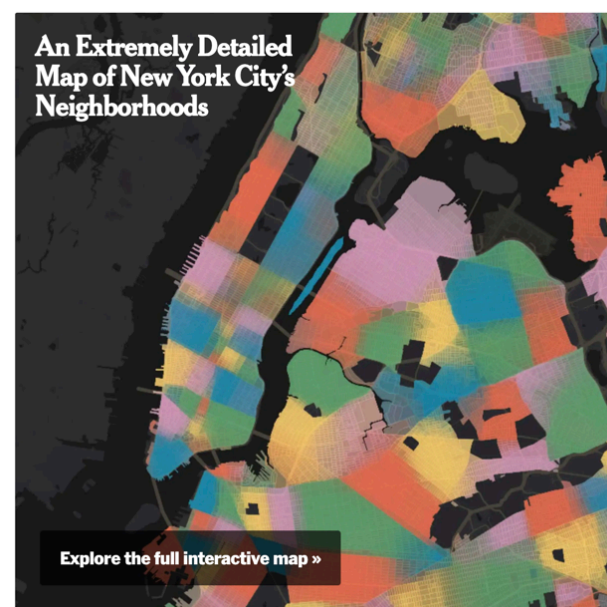
# Neighborhoods

- Jane Jacobs: “In its pure form, the ideal is a neighborhood composed of about 7,000 persons, a unit supposedly of sufficient size to populate an elementary school and to support convenience shopping and a community center. This unit is then further rationalized into smaller groupings of a size scaled to the play and supposed management of children and the chitchat of housewives. Although the "ideal" is seldom literally reproduced, it is the point of departure for nearly all neighborhood renewal plans, for all project building, for much modern zoning, and also for the practice work done by today's architectural-planning students, who will be inflicting their adaptations of it on cities tomorrow.”

## What's in a neighborhood?







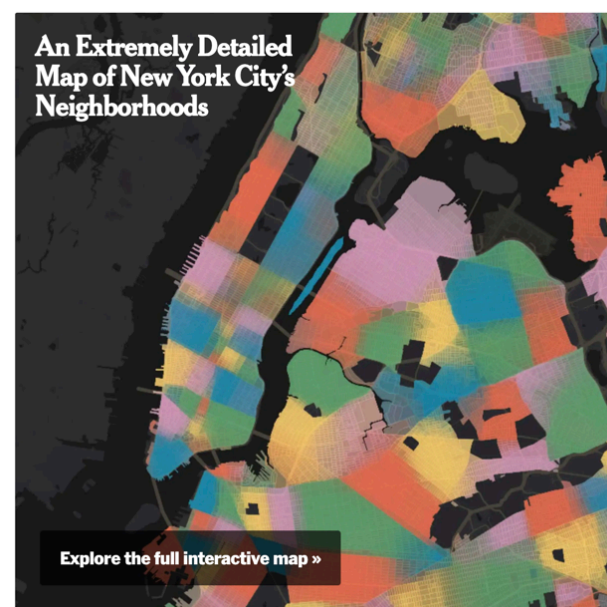
- Neighborhoods are dynamic! Google Maps may feel official, but it allows for crowdsourcing
- Linguists think of languages as “alive” and changing dynamically, geographers think of maps the same way.
- Neighborhood identities can be layered and overlapping, and the physical environment plus the demographic makeup can combine to get a neighborhood to a tipping point of change

LITTLE YEMEN

Google Maps

This neighborhood exists on the map partly because a Yemeni air traffic controller, Yahay Obeid, named it, by reporting that a place was “missing” from Google Maps, and submitting a suggestion for



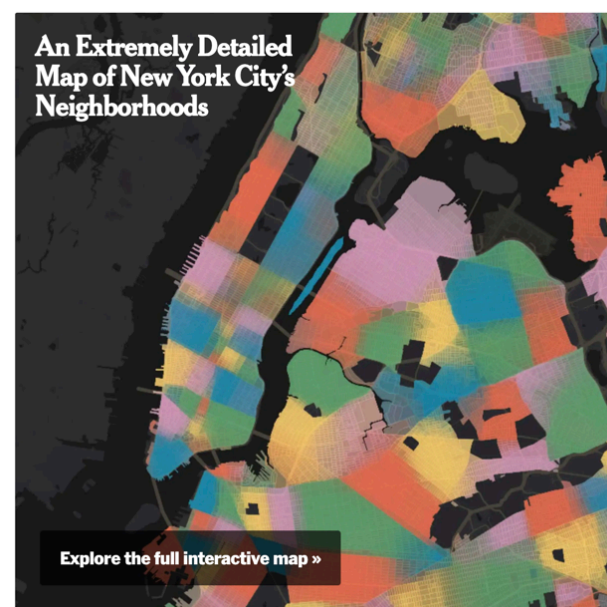


**Sharp borders**  
or the swamp and the expressway

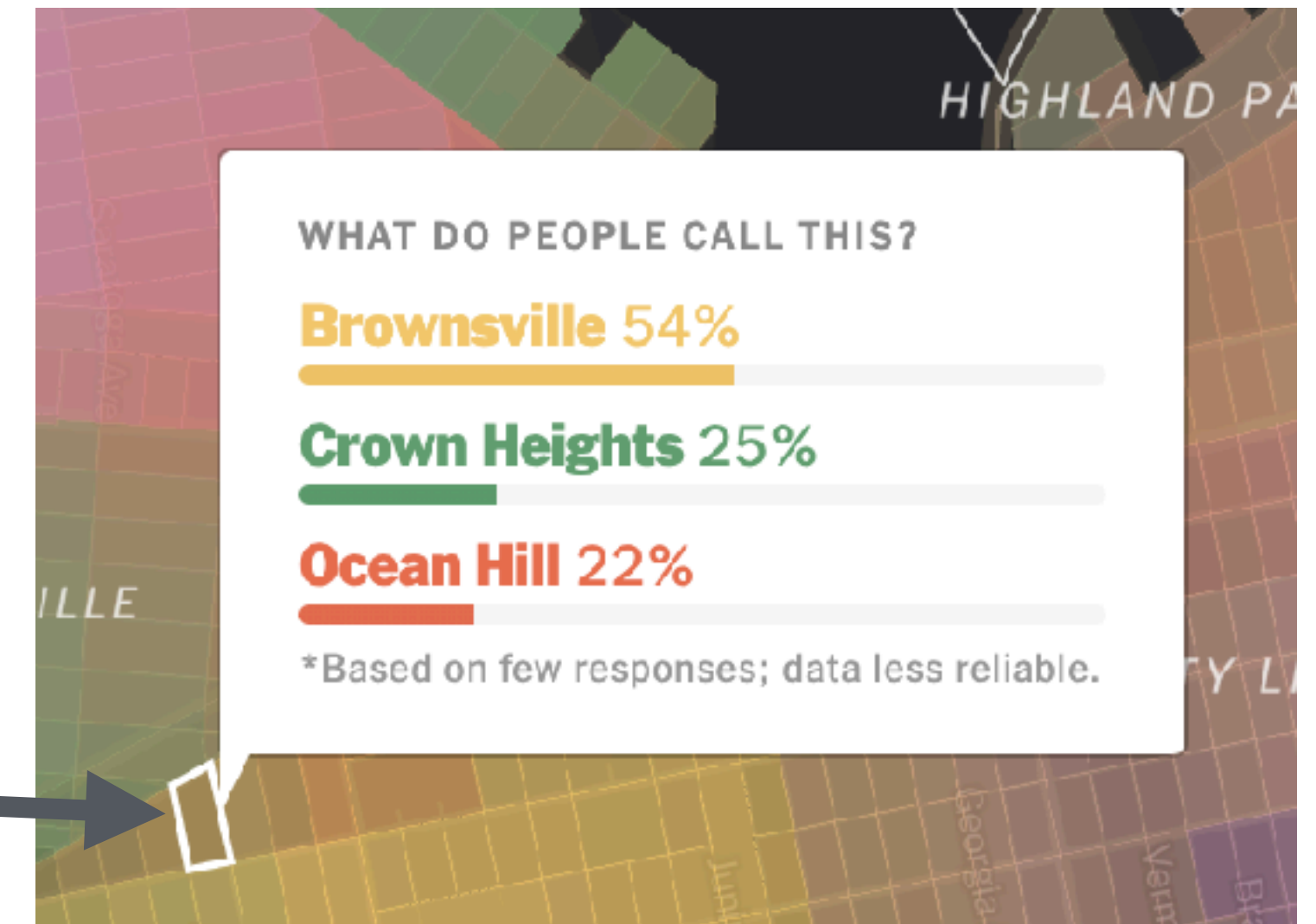
- As we've been discussing, highways can create physical barriers that get hardened into neighborhood lines
- But Williamsburg is an example where the underlying polygon boundaries show that people don't quite agree about the edge of the neighborhood — does it stop at the BQE highway or continue on?



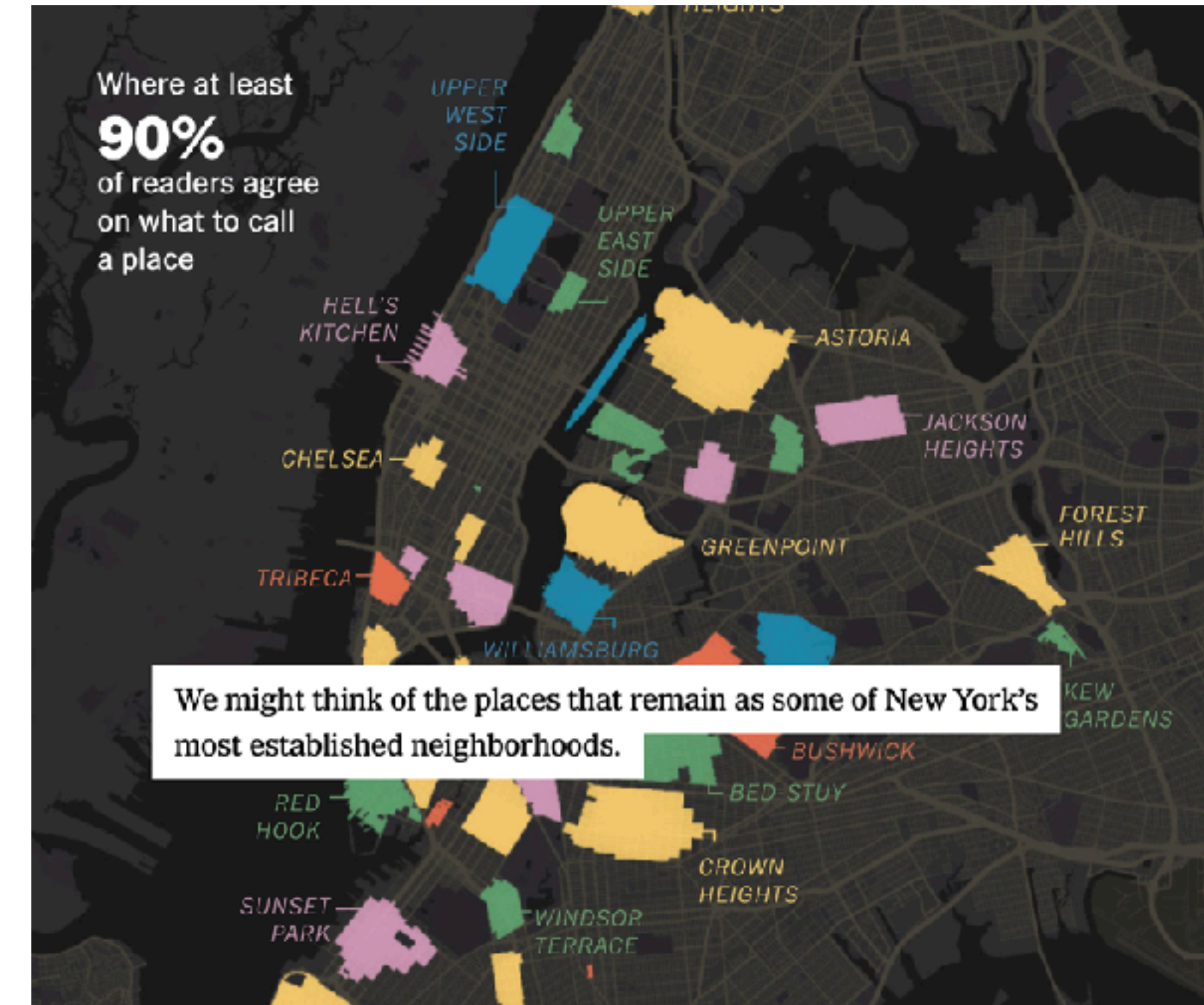




- Some boundaries are fuzzier! And this creates zones of ambiguity, which are like soft edges.



- If you restrict to just the **core** of a neighborhood, where there's at least 90% consensus on the name, then you get a sharper view of some "established" neighborhoods

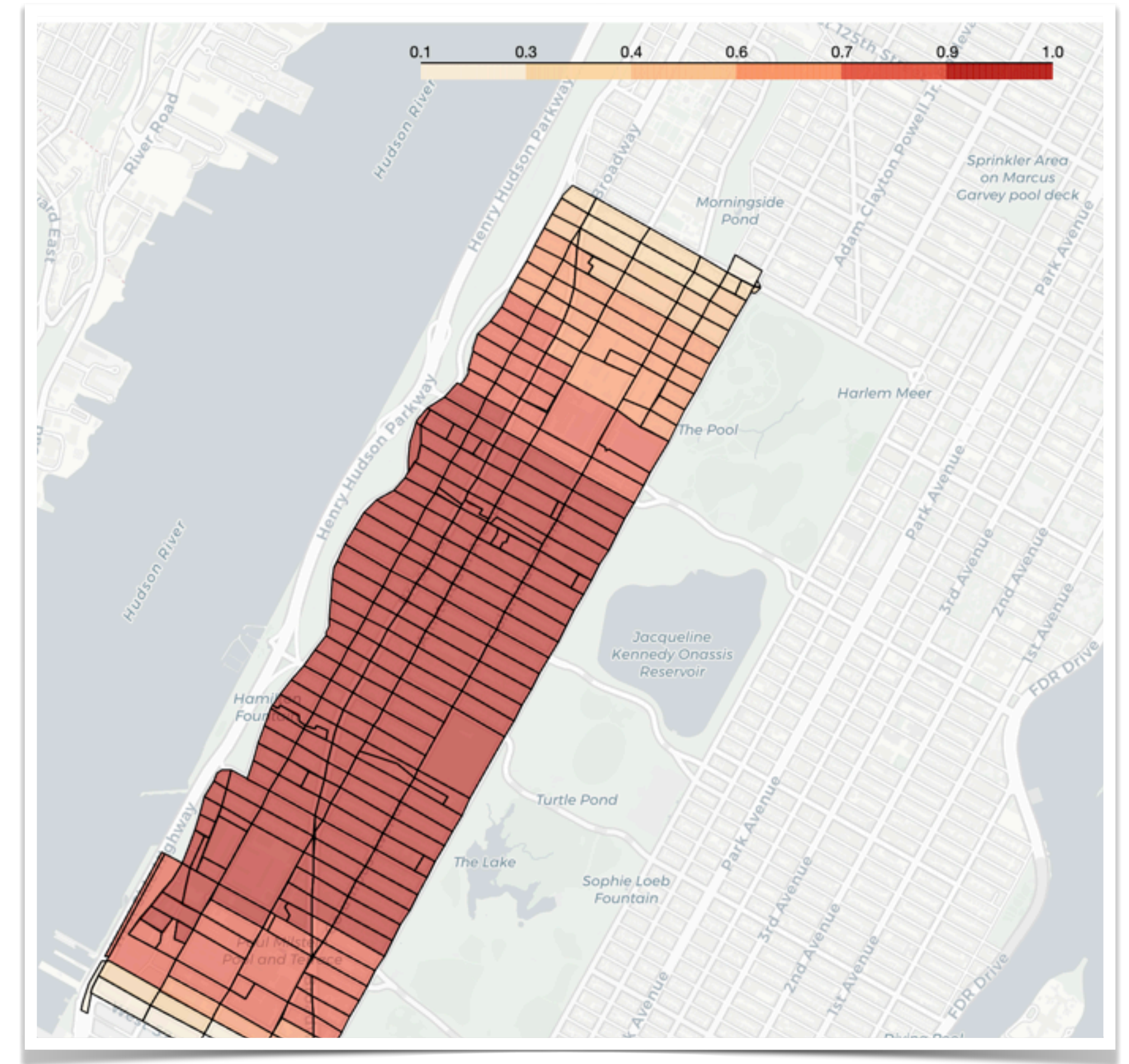




# NYT neighborhood project

- Each block has a weight vector with 379 entries, one for each in the menu of NYC neighborhood names
- [UWS, Tribeca, ....]
- example:

('360850323001008', {'Mariners Harbor': 0.514, 'Graniteville': 0.486}))

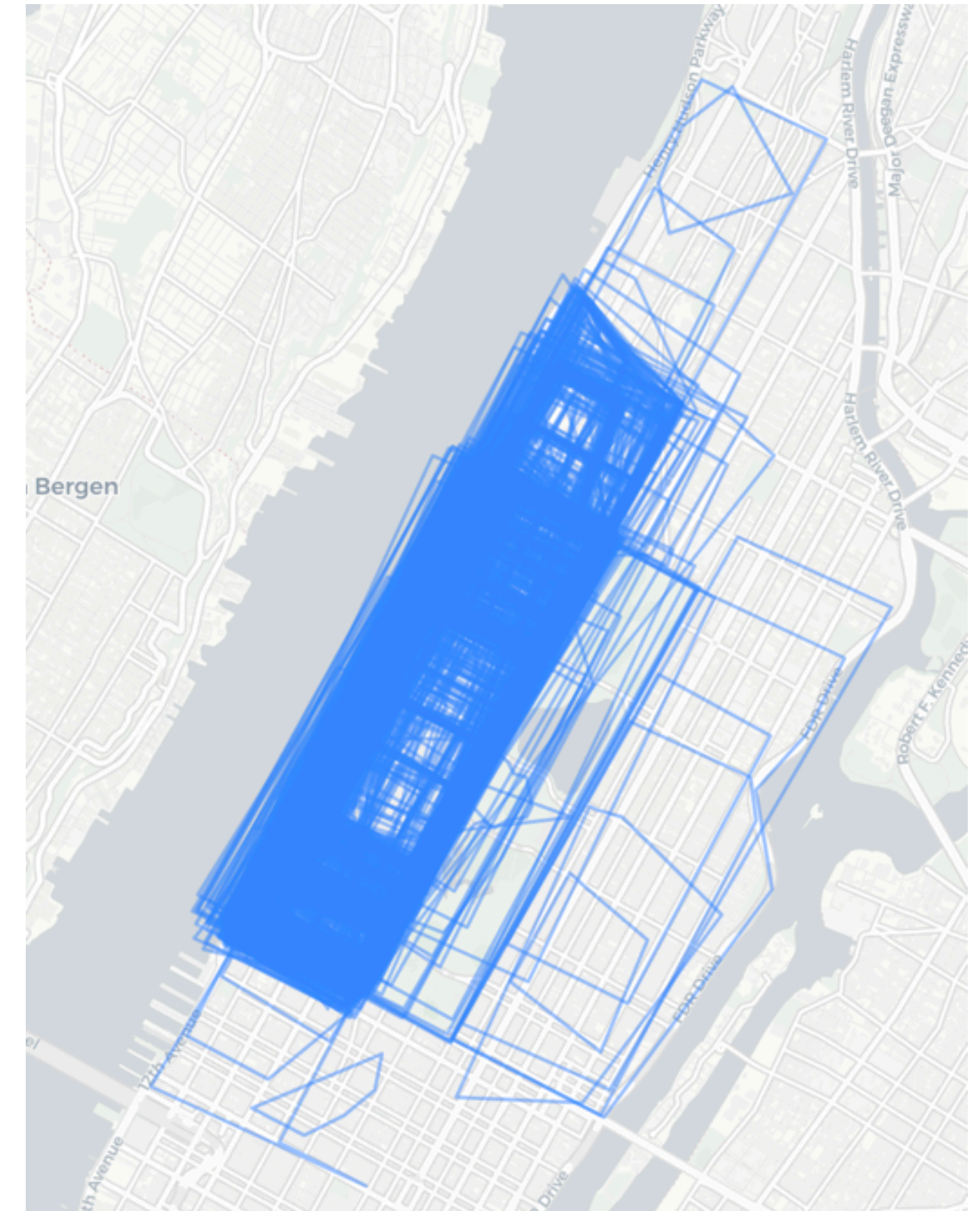


heatmap of "Upper West Side"



- One way to measure fuzziness is to take a block and count how many different neighborhood names it gets
- But this doesn't take into account that some outliers or stragglers could just be confused or wrong or random
- If a block is called UES 99 times and UWS just once, that's suggestive
- We'll use an idea from information theory called **entropy**, specifically Shannon entropy

$$H(w) = - \sum w_i \log w_i$$



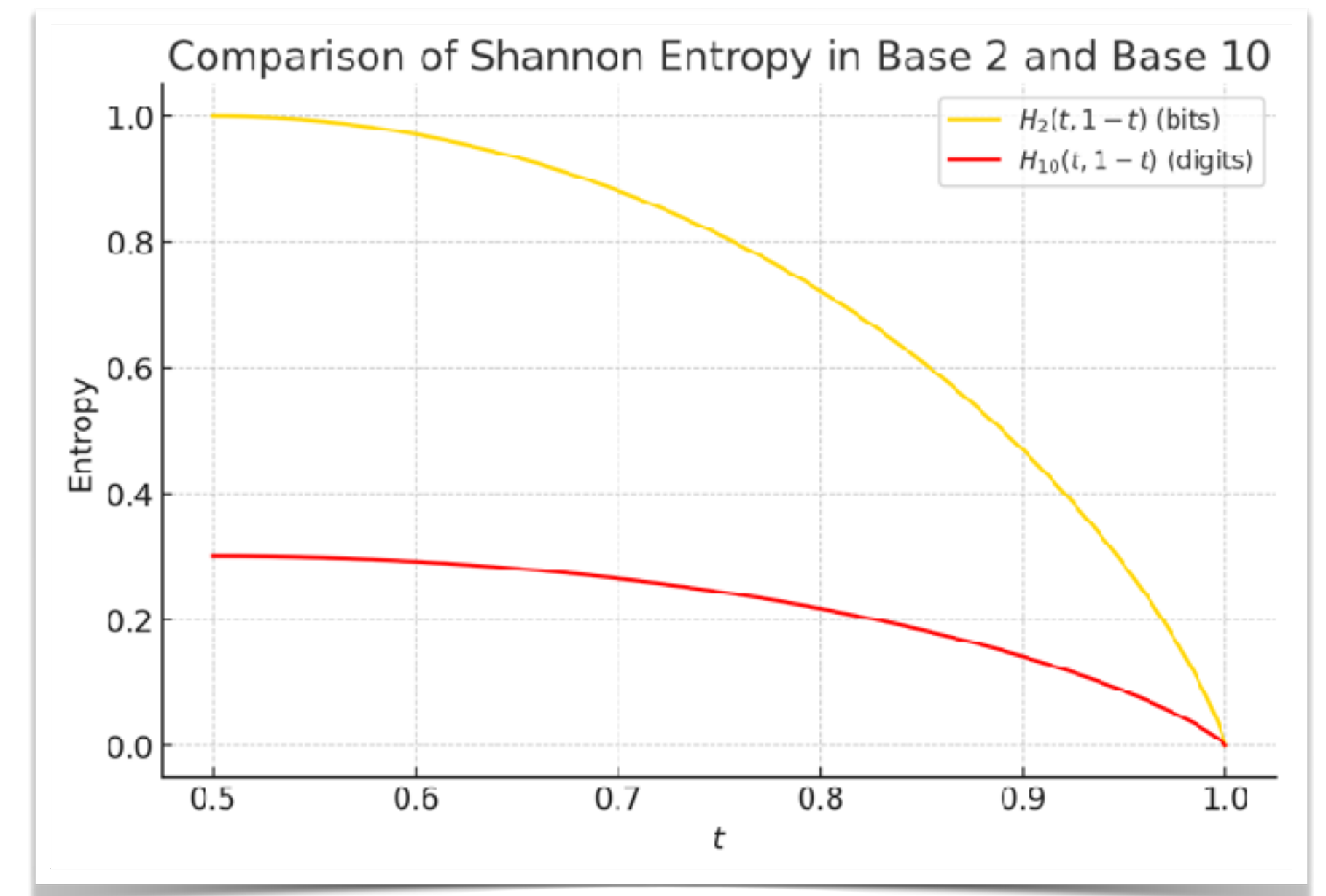
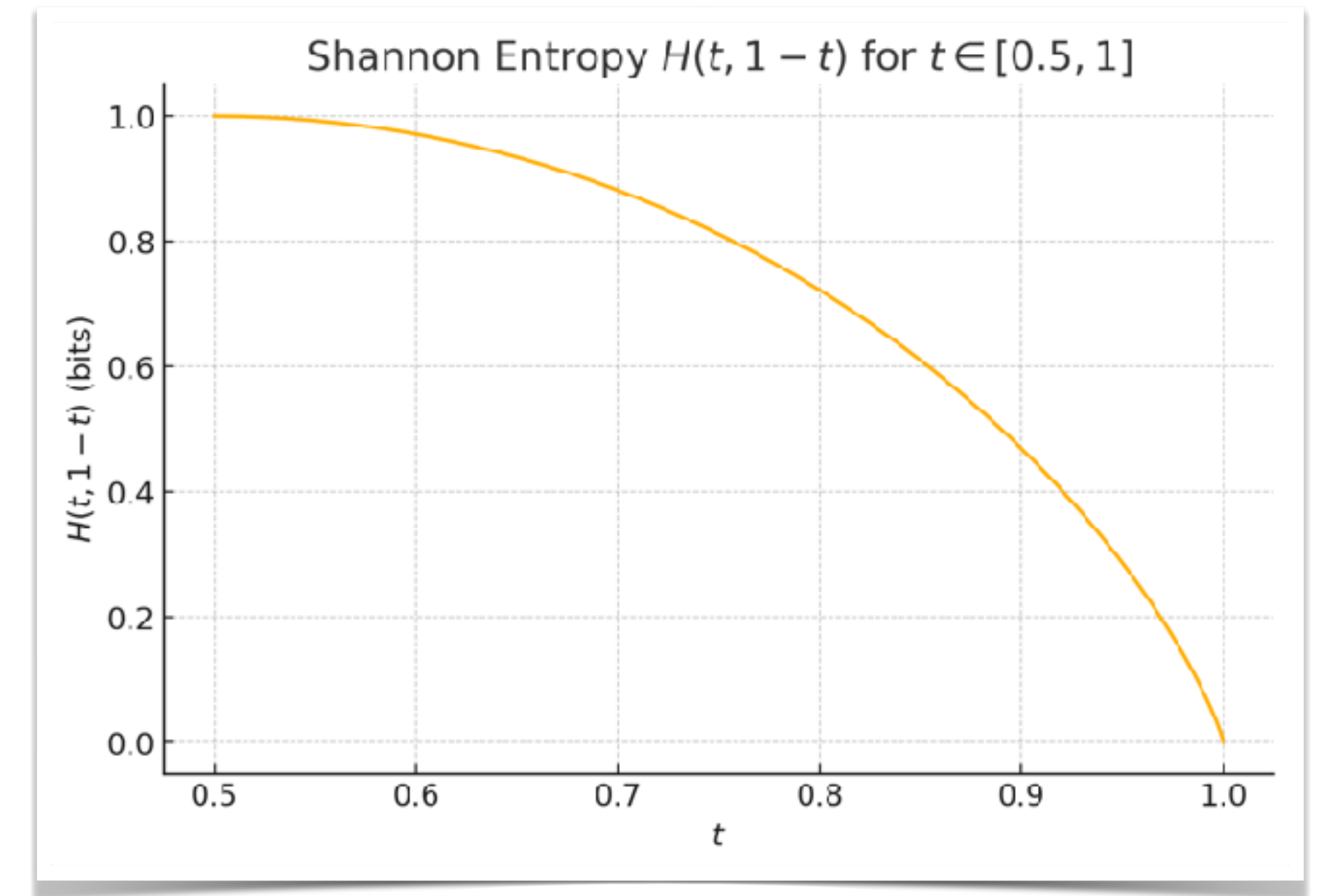
**“Upper West Side”**  
by public mapping



# How entropy works

$$H(w) = - \sum w_i \log w_i$$

- So for (.5, .5) you'd get  $H=1$  (*lots of uncertainty!*)
- $H(.67, .33) \approx .918$
- $H(.9, .1) \approx .469$
- $H(.99, .01) \approx .081$  (*not much uncertainty about label*)
- and this extends to 3 coordinates, or 379
- $H(.33, .33, .33) \approx 1.585$ ;  $H(.5, .4, .1) \approx 1.361$ ;  $H(.8, .1, .1) \approx 0.922$





# How entropy works

$$H(w) = - \sum w_i \log w_i$$

- Minimum number of bits or digits needed to encode something
- Closely connected to the **complexity** of a decision tree
- Appears in physics, cryptography, machine learning, diversity measurements



Figure 1: It takes a lot of bits to give a precise description of a chaotic configuration.

